

Olist E-Commerce Data Engineering Project

Data Cleaning, Database Analysis and Power BI Dashboard

Project Report

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Data Engineering and Analytics Portfolio Project

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Contents

1	Executive Summary	2
2	Introduction	2
3	Project Objectives	2
4	Dataset and Business Context	3
5	Project Architecture	4
6	Data Preparation	4
7	Data Validation	5
8	Database Implementation	6
9	SQL Analysis	6
9.1	Order Volume	6
9.2	Delivery Performance	6
9.3	Customer Distribution	7
10	Power BI Dashboard	7
11	Challenges and Lessons Learned	8
12	Limitations and Future Improvements	8
13	Conclusion	9
14	Tools and Technologies	9
15	References	9

1 Executive Summary

For this project, I worked with the Brazilian Olist e-commerce dataset to understand how order data can be prepared, stored, and used for analysis.

I wanted to build something that covered more than just writing SQL queries or creating charts. So I worked through the different stages of a small data engineering workflow. I started by exploring the raw CSV files, then used Python and pandas to clean and prepare the order data. After that, I loaded the prepared data into SQLite and PostgreSQL and used SQL to investigate order volumes and delivery performance.

I also created a Power BI dashboard to make the results easier to explore. It includes information about customer locations, order trends, delivery performance, and order status.

The project helped me practise working with data from beginning to end. It also gave me a better understanding of how data preparation and analysis fit together in a practical project.

2 Introduction

E-commerce platforms generate a large amount of data every day. Orders, customer locations, payment information, and delivery dates can all be useful when trying to understand how an online business operates.

However, having data does not automatically make it useful. The information needs to be checked, cleaned, organised, and analysed before meaningful conclusions can be drawn.

I chose the Olist Brazilian e-commerce dataset because it contains different types of information about online orders and deliveries. It gave me an opportunity to work with real-world-style data and deal with some of the issues that commonly appear in data projects.

My main focus was the order dataset. I used it to explore order activity and delivery performance, while also practising Python, SQL, database loading, and Power BI.

3 Project Objectives

I set a few practical goals before starting the project:

- Explore the Olist dataset and understand how its tables and fields are organised.

- Use Python and pandas to clean and prepare order records.
- Create additional fields that could help explain delivery performance.
- Check the prepared data for duplicates, missing values, and calculation issues.
- Load the data into SQLite and PostgreSQL.
- Write SQL queries to investigate order activity and delivery-related questions.
- Build a Power BI dashboard that presents the results in a clear way.

These objectives helped me keep the project focused instead of trying to use every dataset and feature available in Olist.

4 Dataset and Business Context

The project uses the Brazilian Olist e-commerce public dataset, available on Kaggle. It contains information about orders placed through an online marketplace, including order status, purchase timestamps, estimated delivery dates, and actual delivery dates.

I focused mainly on the order records because they provided the information I needed for the delivery and order-trend analysis.

Some questions I wanted to explore were:

- How many orders were recorded in the dataset?
- How did order volumes change over time?
- How long did customers wait for their orders?
- How often were orders delivered later than the estimated date?
- Which Brazilian states had the largest number of customers?

The dataset represents historical information. Therefore, the findings in this report describe the records available in the dataset, rather than Olist's current business performance.

5 Project Architecture

I organised the work into a sequence of steps so that each stage had a clear purpose. The raw CSV files were the starting point. Python handled the preparation work, and the cleaned data was then loaded into databases for analysis.

The following diagram shows the general flow I followed.

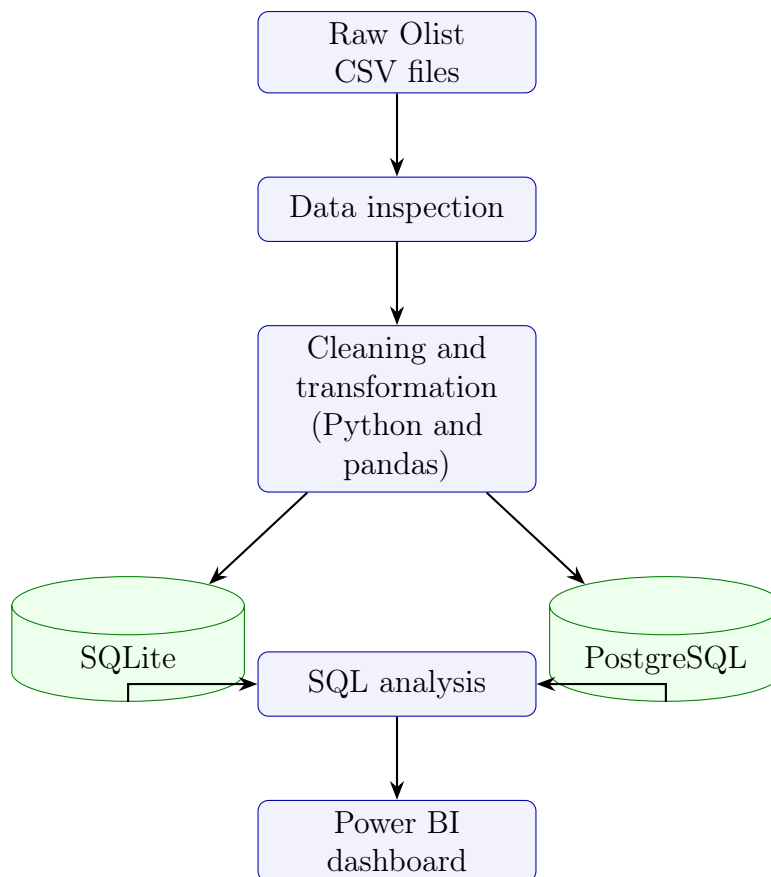


Figure 1: Overview of the project workflow

This is a high-level view of the work. SQLite and PostgreSQL were both used for database practice and analysis. The diagram does not represent an automated pipeline or a live connection between every component.

6 Data Preparation

I started the preparation stage by inspecting the source files and checking their structure. This helped me understand the available columns and identify which fields were relevant to the analysis.

I used Python and pandas for the cleaning and transformation work. Rather than making changes to the original files, I kept the raw data separate from the processed outputs.

The main preparation tasks included:

- Reading and inspecting the order data.
- Checking the records for duplicate order identifiers.
- Working with date and timestamp fields.
- Preparing delivery-related fields for analysis.
- Saving the cleaned and transformed data separately.

The cleaned dataset contained 99,441 order records and 8 columns. I then created a transformed version with 12 columns, including additional fields used in the delivery analysis.

Keeping the cleaning and transformation stages separate made it easier to understand what had changed and to check the results.

7 Data Validation

After preparing the data, I ran validation checks to catch issues that could affect the analysis.

I checked for duplicate order identifiers and reviewed the delivery-related calculations. I also looked for negative delivery durations and records with missing delivery information.

The validation results showed no duplicate order identifiers in the cleaning check. I also found no negative delivery durations or calculation mismatches in the checks I performed.

There were 2,965 records with missing delivery fields. These records needed to be treated carefully because a missing delivery date does not necessarily mean that an order was delivered late. Some orders may not have had a completed delivery at the time represented by the dataset.

This step was useful because it helped me understand the limitations of the data before using it in SQL queries and dashboard visuals.

8 Database Implementation

I used both SQLite and PostgreSQL during the project. Working with two database systems gave me the chance to practise loading prepared data and running queries in different environments.

For SQLite, I created a database file named `olist.db` and loaded the prepared order data into an `orders` table. I checked that the table contained 99,441 rows.

I also loaded the data into PostgreSQL, using a database named `olist_db`. This gave me a database environment for running SQL queries and exploring the prepared records.

The database stage was an important part of the workflow. It helped me move from working with CSV files in Python to querying structured tables directly.

9 SQL Analysis

Once the data was available in PostgreSQL, I wrote SQL queries to answer questions about order activity and delivery performance.

I focused on a few areas rather than trying to create a large collection of unrelated queries. These included overall order counts, monthly order volumes, delivery durations, and delivery delays.

9.1 Order Volume

The dataset contained 99,441 order records. When I grouped the records by month, November 2017 had the highest monthly order volume in the analysis, with 7,544 orders.

This gave me a useful starting point for looking at changes in order activity over time.

9.2 Delivery Performance

For the delivery analysis, I compared actual delivery dates with estimated delivery dates where the required fields were available.

The average delivery duration was approximately 12.56 days. The average estimated delivery duration was approximately 23.74 days. Based on the calculation used in the project, the average difference was around -11.18 days.

A negative difference in this calculation means that the actual delivery date was earlier

than the estimated date on average.

The analysis also identified 7,826 late deliveries and 88,644 deliveries classified as on time or early by the query used. These categories should be interpreted according to the query's filtering and date-field rules, rather than assumed to cover every order record.

9.3 Customer Distribution

I also examined customer distribution by Brazilian state. This made it possible to see where customers were concentrated in the available dataset.

The state-level analysis was later included in the Power BI dashboard so that the distribution could be explored visually.

10 Power BI Dashboard

After completing the main SQL analysis, I used Power BI to bring some of the findings together in one place.

I wanted the dashboard to be easy to read without requiring someone to run SQL queries or inspect the source files. I therefore included visuals for several aspects of the dataset.

The dashboard contains:

- Customer counts by Brazilian state.
- Delivery performance information.
- Delivery delay analysis.
- Order status distribution.
- Order trends over time.
- Order activity by weekday.
- An order-volume gauge.

The visuals provide different ways to look at the same dataset. For example, the monthly trend helps show how order activity changed over time, while the delivery visuals focus on how orders performed against their estimated dates.

The dashboard is intended as an analytical view of the historical data. It is not a live monitoring system.

11 Challenges and Lessons Learned

One of the first things I noticed was that the raw data needed some preparation before it could be used confidently. Date fields, missing values, and the structure of the source files all needed attention.

Another part that required care was the delivery analysis. It was important to distinguish between orders that were delivered late and records that did not have enough delivery information for a comparison. Treating all missing values as late deliveries would have produced misleading results.

I also had to make sure that the data loaded into the databases matched the prepared dataset. Checking row counts and validating calculations helped me catch potential problems before moving on to the analysis.

Working through these steps gave me a clearer picture of the work that happens before a dashboard is created. The visualisation is only one part of the process; the quality and structure of the underlying data matter just as much.

12 Limitations and Future Improvements

The project covers a useful set of data preparation and analysis tasks, but there are areas I would improve if I continued developing it.

First, the analysis focuses mainly on order records and delivery performance. The wider Olist dataset includes other information, such as payments, products, sellers, and reviews. Combining more of these datasets could help answer broader business questions.

Second, the workflow is currently made up of individual scripts and database-loading steps. I could make it more repeatable by introducing an orchestration tool such as Apache Airflow and adding scheduled execution.

I would also improve the validation process by adding more automated checks and clearer reporting of records that fail those checks.

Finally, the dashboard could be extended with additional filters and measures. This would make it easier to investigate particular time periods, order statuses, or customer groups.

These are possible next steps rather than features already implemented in the current project.

13 Conclusion

This project gave me the opportunity to work through a practical data engineering and analytics workflow using the Olist e-commerce dataset.

I began by exploring the raw data and preparing the order records with Python and pandas. I then validated the results, loaded the data into SQLite and PostgreSQL, and used SQL to examine order volumes and delivery performance. Finally, I created a Power BI dashboard to present some of the findings visually.

The main thing I learned is that useful analysis depends on more than writing queries or building charts. Understanding the source data, preparing it carefully, and checking the results are all important parts of the process.

I see this project as a practical example of the skills I have been developing in Python, SQL, databases, and data visualisation. It also gives me a foundation to build on with more advanced data engineering projects.

14 Tools and Technologies

The following tools were used in the project:

- **Python and pandas:** Used to inspect, clean, and transform the order data.
- **SQLite:** Used to store and query the prepared data in a local database.
- **PostgreSQL:** Used for database loading and SQL analysis.
- **SQL:** Used to investigate order volume, delivery performance, and customer distribution.
- **Power BI:** Used to create the dashboard and present the analysis visually.
- **Git and GitHub:** Used to manage and publish the project files.

15 References

- Olist Brazilian E-Commerce Public Dataset. Available on Kaggle: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>
- Python Documentation: <https://docs.python.org/3/>

- PostgreSQL Documentation: <https://www.postgresql.org/docs/>
- Microsoft Power BI Documentation: <https://learn.microsoft.com/power-bi/>